TRANSFINITTE · HARDWARE TRACK · TEAM HANDOUT

# HridayMitra: summary of what we've done, and the full component list

**Monitoring only, not a diagnosis tool. Nothing is built or tested yet.**

## Part 1. Summary of what we've done

### 1. Understood the problem statement

- Early warning of worsening heart failure at home, days before hospital care, for an elderly patient in an Indian home.
- Rules: under 2 minutes a day (no all-day patch), local language, power cuts and weak internet, home-unit parts under ₹5,000.
- Judging: sensing creativity 20, ease and long-term use 20, software 20, hardware and simulation 15, science and honest testing 15, cost, India fit and pitch 10.
- Deliverables: user journey, block diagram and schematic, parts list, simulation proof, code, 20+ synthetic patients × 30 days with four metrics, GitHub, 2-page report, 5-minute pitch.

### 2. Researched existing solutions (four tracks)

- Weight rules alone catch about 23–35% of events with about 4 false alerts per patient-year.
- A scale adding foot impedance caught 70% in one company-linked study. Implants set the ceiling at about 70% with 34 days of warning.
- Voice analysis reported 71–85% sensitivity (company-linked). Large telemonitoring trials saw use fall to about 55%, and outcomes depended on a team acting on alerts.
- India: eSanjeevani and ASHA workers exist as a response network. 2G still runs on Airtel, Vi and BSNL but not Jio. Dozee (hospital-focused) is the closest Indian competitor.

### 3. Options explored, and why most were rejected

| Option | Verdict | Reason |
|---|---|---|
| Cot or bed sensor | Rejected | Other sleepers, bedding and pillows interfere; no heart failure data |
| Wrist band + dock | Rejected | About half of users stop wearing bands; no weight or fluid signal |
| Ankle band, toe ring | Rejected | Pressure and circulation risk on swollen feet; weak signals; not everyone wears them |
| Pill-box vitals | Not main idea | Creative, but no weight or fluid signal. The pill reminder survives as a software feature. |
| Chest puck, pill-tray hardware, multi-frequency chip | Future add-ons | Cost and scope |

### 4. The final product: HridayMitra

- **Check-in station:** a foldable all-in-one stool (seat, footrest with 4 foot electrodes, grab handle with 4 hand electrodes and a finger pulse and $SpO_2$ clip, 4 load cells, speaker, 3 lights, 2 buttons and SOS, SIM800L modem, battery). About 75–90 s each morning.
- **Locket:** a chest pendant with a motion sensor for night breathing, shallow-breathing index, body angle, activity, falls and SOS. Part of the base product.
- **Alert engine** runs on the device: personal baseline, amber/red levels, danger rules, a spoken reason for every alert, cause tags.
- **Locked choices:** trunk impedance via the handle, 2G SIM800L for the build (4G is the product upgrade), ECG skipped for now, foldable stool, locket in base.

### 5. What makes it different

- Fluid and heart-pumping from the same electrodes (the pumping part is a stretch goal).
- Two-witness alerts with a confidence score and "no reading" instead of guessing.

- A closed care loop: reply-by-SMS (1, 2, 3), escalation timers, an ASHA card (wording needs cardiologist approval).
- **Fluid-shift index (new, approved):** a 20-second foot-only reading each evening, compared with the morning one, to show how much fluid moved from the legs toward the chest overnight. The mechanism is documented in sleep-apnea research (PMC3607865). It is a hypothesis for heart failure at home: I found no study of it.
- **Two-frequency leg method (new, approved):** measure at about 5 kHz and 50 kHz with the classic ESP32's own signal generator, at no extra hardware cost.

### 6. Honest assessment

- No 100% accuracy is possible. Fluid in the lungs is estimated from several signals, not measured directly. The legs are measured as a trend against the patient's own baseline.
- My ideation score estimate: about 66/100 now. A realistic target is 85–90 if the schematic, simulation, code and honest results are built.
- Startup framing: the kit at near cost plus a monthly care-loop service, bought by families, hospitals and government programmes. All numbers are hypotheses.

### 7. Files made so far

- **HridayMitra ideation report.pdf** (20 pages: full research)
- **HridayMitra final design.pdf** (6 pages, now superseded)
- **HridayMitra product brief.pdf** (the main product document; updated with the approved changes)
- This handout. The shared team page (a link from earlier) still shows the old wrist-band plan.

### 8. Not done yet

Everything that is scored on proof: the block diagram and schematic, the LTspice and Wokwi simulations, the code, the 200+ synthetic patients and results, the BIDMC heart-rate test, the GitHub page, the report and the pitch.

## Part 2. Complete component list (every part, including the small ones)

**Read this first.** Two cost columns: **Low** is the cheapest plausible price and **Planning** is a cautious price. The earlier accepted price list came to about **₹4,765** (station ₹4,032, locket ₹733). This itemized list adds every small part (power, protection, mechanical, board and wiring), so it totals **₹4,125 at low prices and ₹6,209 at planning prices**. The ₹5,000 limit is met only if real quotes land near the low column. Until P1 gets real quotes, treat the limit as **unproven**. Only items marked "Verified range" or "Weak listing" were web-searched. Everything else is my estimate.

| # | Part | Qty | What it does | Low ₹ | Plan ₹ | Price status |
|---|---|---|---|---|---|---|
| **A. Brain and power** | | | | **750** | **1,040** | |
| 1 | ESP32 DevKit, classic WROOM-32 (needs the built-in DAC) | 1 | Main processor, impedance signal generator, Bluetooth to locket | 450 | 600 | Estimate (only S3 boards priced, ₹799–1,299) |
| 2 | 18650 Li-ion cell, 2000–2600 mAh | 1 | Battery for power cuts | 100 | 130 | Estimate |
| 3 | 18650 holder | 1 | Holds the cell | 15 | 20 | Estimate |
| 4 | TP4056 charge module with protection | 1 | Charges and protects the cell | 30 | 45 | Estimate |
| 5 | 5 V boost/buck module, 2 A or more | 1 | Stable supply for ESP32, DFPlayer, modem bursts | 40 | 70 | Estimate |
| 6 | 5 V 2 A mains adapter and cable | 1 | Charging | 100 | 150 | Estimate |
| 7 | Power switch, 470–1000 µF capacitor | 1 set | On/off, modem burst current | 15 | 25 | Estimate |
| **B. Weight** | | | | **540** | **695** | |
| 8 | 50 kg load cell | 4 | One under each frame foot | 480 | 600 | Verified range (₹136–220 per set with HX711) |
| 9 | HX711 24-bit ADC board | 1 | Reads the load cells | 40 | 55 | Verified range (₹40–59) |
| 10 | 4-cell bridge board and wiring | 1 | Combines the four cells | 20 | 40 | Estimate |
| **C. Impedance front end (the custom circuit)** | | | | **230** | **425** | |
| 11 | Dual op-amp, rail-to-rail, 3 MHz or more | 2 | Current source and filters | 40 | 70 | Estimate |
| 12 | Instrumentation amplifier (AD620 or INA126 class) | 1 | Reads the small voltage across the inner electrodes | 100 | 200 | Estimate |
| 13 | Analog multiplexer (CD4051 or 74HC4052) | 2 | Switches between leg and chest paths | 30 | 50 | Estimate |
| 14 | Precision resistors incl. 100 Ω and 1 kΩ reference | 15 | Current setting, reference check | 25 | 45 | Estimate |
| 15 | Capacitors, assorted ceramic and film | 20 | DC blocking, filtering | 20 | 35 | Estimate |
| 16 | Schottky diodes and TVS (ESD) diodes | 8 | Rectifier and electrode protection | 15 | 25 | Estimate |
| **D. Electrodes and cabling** | | | | **240** | **390** | |
| 17 | Stainless-steel foot plates | 4 | 2 per foot: one injects, one senses | 100 | 160 | Estimate |
| 18 | Stainless-steel hand grips with insulating split | 4 | 2 per hand | 100 | 160 | Estimate |
| 19 | Shielded multi-core cable | 1 | Electrodes to the electronics box | 40 | 70 | Estimate |
| **E. Pulse and $SpO_2$** | | | | **106** | **175** | |
| 20 | MAX30102 module | 1 | Finger pulse and oxygen | 96 | 150 | Weak listing (₹96) |
| 21 | Finger clip holder | 1 | Holds the finger in place | 10 | 25 | Estimate |
| **F. Voice and interface** | | | | **769** | **974** | |
| 22 | DFPlayer Mini MP3 module | 1 | Plays voice prompts, drives the speaker | 260 | 330 | Verified range (₹260–349) |
| 23 | Speaker, 4 Ω 3 W | 1 | Loud voice for hard-of-hearing users | 249 | 249 | Verified range (Probots) |
| 24 | microSD card, 2–4 GB | 1 | Voice files in local languages | 120 | 180 | Estimate |

| | | | | | | |
|---|---|---|---|---|---|---|
| 25 | Large push buttons: green, red, SOS | 3 | Answers and emergency | 120 | 180 | Estimate |
| 26 | LEDs green, amber, red, with resistors | 3 | Status lights | 20 | 35 | Estimate |
| **G. Connectivity** | | | | **150** | **350** | |
| 27 | SIM800L GSM (2G) module with antenna | 1 | SMS and voice calls. No Jio. SIM and recharge are a running cost, not counted. | 150 | 350 | Verified range (₹150–448) |
| **H. Mechanical** | | | | **590** | **870** | |
| 28 | Foldable stool frame: seat, footrest, handle post, hinges | 1 | The all-in-one body | 400 | 550 | Estimate |
| 29 | Load-cell mounting plates and bolts | 4 | Hold the cells under the frame feet | 60 | 100 | Estimate |
| 30 | Electronics enclosure under the seat | 1 | Protects the board | 60 | 100 | Estimate |
| 31 | Screws, grommets, heat-shrink, cable ties | 1 set | Assembly | 40 | 70 | Estimate |
| 32 | Anti-slip footrest mat | 1 | Safety | 30 | 50 | Estimate |
| **I. Board and wiring** | | | | **140** | **310** | |
| 33 | Prototype PCB or perfboard | 1 | Holds the circuit | 80 | 150 | Estimate |
| 34 | Jumper wires, headers, screw terminals | 1 set | Connections | 60 | 120 | Estimate |
| 35 | Solder and consumables | 1 set | Assembly | 0 | 40 | Estimate |
| **J. Locket dock (on the station)** | | | | **40** | **70** | |
| 36 | Pogo pins, magnets, small contact board | 1 set | Holds and charges the locket | 40 | 70 | Estimate |
| **Station total** | | | | **3,555** | **5,299** | |
| **K. Locket** | | | | **570** | **910** | |
| 37 | ESP32-C3 SuperMini | 1 | Processor and Bluetooth | 250 | 350 | Estimate |
| 38 | Motion sensor (MPU6050 or LIS3DH) | 1 | Breathing, body angle, activity, falls | 80 | 150 | Estimate |
| 39 | LiPo battery, 300–400 mAh, with protection | 1 | Power | 120 | 180 | Estimate |
| 40 | USB-C or TP4056 mini charge module | 1 | Charging | 20 | 40 | Estimate |
| 41 | SOS micro-switch | 1 | Emergency button | 10 | 20 | Estimate |
| 42 | Case and cord (3D-printed or pendant case) | 1 | The wearable shell | 60 | 120 | Estimate |
| 43 | Resistors, capacitors, pogo contacts | 1 set | Support parts | 30 | 50 | Estimate |
| **Locket total** | | | | **570** | **910** | |
| **Grand total (station + locket)** | | | | **4,125** | **6,209** | **Limit: 5,000** |

### Ways to cut cost if the total runs over (all savings are rough guesses)

| Option | Saves ₹ | Cost of doing it |
|---|---|---|
| Hacked cheap bathroom scale instead of separate load cells, bridge board, mounting plates and part of the frame | about 400–700 | Price and fit unverified; conflicts with the locked foldable frame |
| Cheaper frame: plastic stool plus steel strips | about 100–200 | Less polished, less foldable |
| Voice from ESP32 flash with an I2S amplifier instead of DFPlayer and SD card | about 0–150 | More code, fewer languages |
| Bulk passives and cables from wholesale markets | about 100–200 | Needs a shopping trip or IndiaMART |
| Skip the finger sensor | 150–175 | Loses $SpO_2$ and pulse. Not recommended. |

### Tools and software (not counted in parts)

- **Tools the team probably has:** soldering iron, multimeter, wire strippers, screwdrivers, glue gun or drill for the frame.
- **Free software:** LTspice and KiCad or EasyEDA for the circuit, Wokwi for firmware, Python with NeuroKit2 and SciPy for signals and virtual patients, GitHub for the public page.

- **Running costs, not parts:** a SIM card and recharge for SMS and calls.

Check every part before buying, especially: a classic ESP32 (not S3, which has no signal generator), the instrumentation amplifier, load-cell sizes and the finger sensor. Current injected into the body must stay within safe limits (IEC 60601-1), with an "impedance off" setting for people with pacemakers.